

2082

Bachelor of Science in Computer Science and Information Technology

Course Title: **Data Structure and Algorithms**

Code No: **CSC211** Semester: **3rd Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Distinguish between data type and abstract data type. List any three characteristics of algorithm. Evaluate the postfix expression $33 + 2 * 40 -$, using stack.
2. Define height, depth and level of the tree. Find the minimum spanning tree of the following weighted graph using Kruskal and Prim's algorithm.
3. Define queue as ADT and discuss about primitive operations in queue. Trace the output generated by following recursive function for the statement `TOH(4,1,3).TOH(n, s, d){if (n == 1){printf(" move the top disk from rod %d to rod %d",s,d)}else{TOH (n- 1, s, 6- s- d)printf(" move the top disk from rod %d to rod %d",s,d)TOH (n- 1, 6- s- d, d)}}`

Section B

Attempt any Eight questions[8×5=40]

4. Why do we need algorithms for searching? Trace the binary search operation for the key = 6, in the set {3, 6, 8, 12, 67, 89, 90, 100}
5. How do you implement single linked list using stack? Explain.
6. Trace the merge sort algorithm for the input {8, 7, -4, 19, 3, 45, 12}.
7. Explain about any three hash collision resolution techniques.
8. Describe the process of deleting the node at start, end and specified position of singly linked list.
9. Write the algorithm of Quick sort with its efficiency analysis.
10. Why do we always prefer the worst case for algorithm analysis? Describe the insertion and deletion operation in BST.
11. Define overflow and underflow. Discuss the advantages and limitations of circular queue.
12. What are the advantages of circular queue over linear queue? Sort the array {4, 7, 3, 2, 1} using insertion sort.